Process and Instrumentation Design Project 2026

Contributes 20% to the final mark for the unit

To be submitted to the LMS by 11:59 pm Wednesday 13th May 2026

"Scientists study the world as it is; engineers create the world that has never been," Theodore van Kármán – Aerospace engineer and physicist.

"As engineers, we were going to be in a position to change the world – not just study it," – Henry Petroski – Engineer.

"Engineering is the application of science for human benefit." – J. M. Prausnitz – Chemical Engineer.

"We are continually faced by great opportunities brilliantly disguised as insoluble problems," – Lee Iacocca – Engineer and Automobile Executive.

Why learn to work in groups?

The following is an extract from the report titled "ENGINEERING FUTURES 2035: A scoping study"[1]

> The AICD view is that specialists will be working increasingly in cross-functional teams and higher order soft skills such as empathy, professional ethics and emotional judgment are likely to be increasingly valued. This view is reinforced in the 2018 report from The Association of Independent Schools in NSW which analyses interviews held with 15 CEOs, managing directors and leaders to explore the CEO's views on how technology-driven disruption and associated economic and societal changes were affecting their organisations now, anticipated future impacts, and what this might mean for education. Many of the global trends identified by other reports were also cited by the CEOs from a range of organisations in Australia who were interviewed for this report, some of whom say:
>
> *'Pure technical ability will no longer be sufficient; creative and interpersonal skills will be required even in domains which were once predominantly technical'*
>
> Domain expertise will still be valued, although demands for specific skills are likely to fluctuate and changing skills requirements will lead to greater diversity in the workplace. The rapid pace of change means that some skills will go out of date quickly. Continuing lifelong learning will be essential. Adaptability, flexibility and

[1] Peter L Lee, Chair, Steering Committee, *ENGINEERING FUTURES 2035: A scoping study*, Australian Academy of Engineering Deans, 2019.

resilience will be increasingly important. Those working in start-ups and small companies in particular will be:

'asked to do things they have no experience in and that things can – and will – change dramatically from day to day'

Assessed outcomes

- Ability to find and discriminate information from data bases, such as libraries, technical and academic articles and the internet, relevant to the questions
- Ability to write concisely, clearly and logically
- Ability to interpret and put information together in a logical way, and to understand the implications.
- Ability to work in a team to achieve a common goal
- Ability to work to international standards
- Ability to write readable and maintainable programs for PLCs

Background

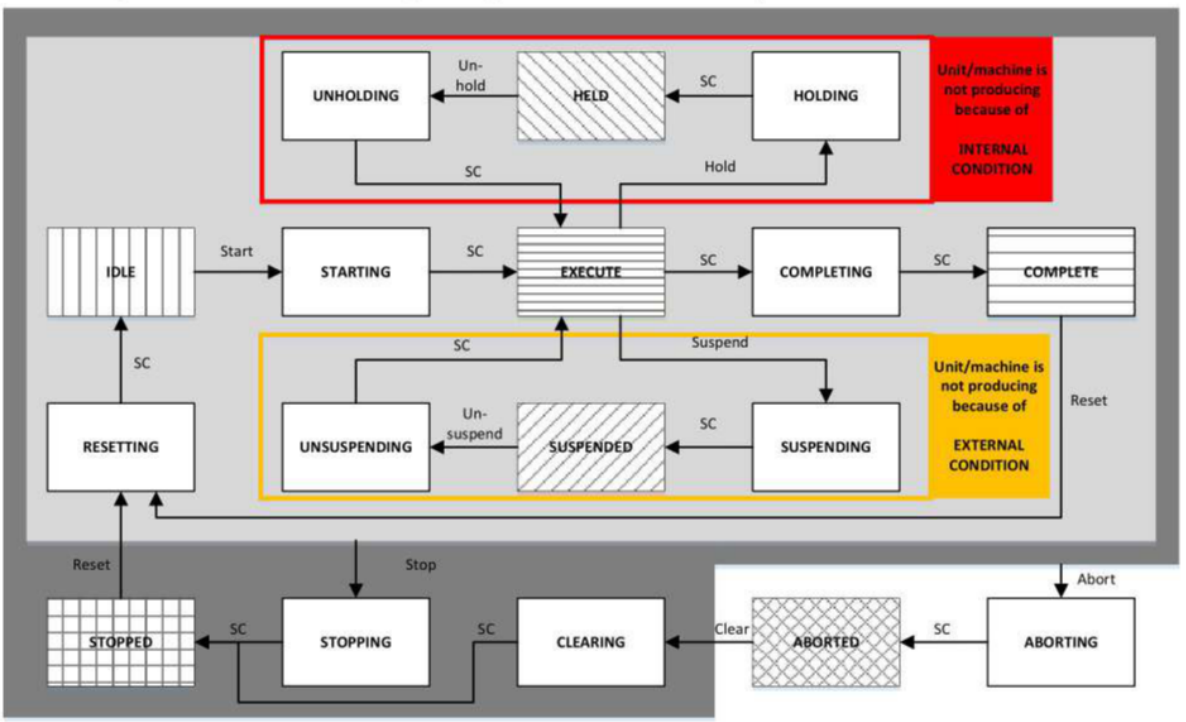
- A manufacturer produces two types of products: A red container containing Fluid A, a viscous fluid, and blue container, also a viscous fluid, containing Fluid B.
- The structure of the factory is that each step of the process is performed by a work-cell. Robot arms move the output of one cell to the next cell in the sequence under a command from the next cell.
- The batch process consists of a cell that produces empty red containers, a cell that produces empty blue containers, a cell that fills the containers depending on the colour of the container with the appropriate fluid and a cell that seals the containers with a common lid.
- The filling cell is supplied with containers by a robot which places a container on the conveyor belt of the filling station.
- A separate robot collects the filled containers at the end of the filling station's conveyor belt.
- The manufacturer uses the PackML standard in the factory.
- You have been contracted to write the program to control the filling work cell. The program must comply with the full PackML standard.
- The customer has fitted an acoustic level sensor (High when Full), proximity sensor for the container at the input. (High when a container is present), and separate solenoids that operate the filling of fluid A and fluid B (High opens the filling injector).

- The cell has a conveyor belt to move the container to the appropriate filling location and to shift it to the region where the robot arm picks up the container for the next cell. (The conveyor belt is turned on with a High signal.)
- Not all of the necessary items have been included and you are free to add sensors and drivers as you see fit.
- You should also consider fault situations and what should be done about them. That is what triggers an Abort or Stop etc

Deliverables

The following deliverables are required:

1. State diagram of the PackML (ANSI/ISA TR88.02-2015) structure:



2. A Structured Text program written on the CODESYS platform for each state of the PackML structure. The execute state is where your program for the filling of the containers resides;
3. A Sequential Function Chart written on the CODESYS platform representing the PackML structure using the structured text programs of each state in each step;
4. A Visualization written on the CODESYS platform to demonstrate that the program functions correctly. The Visualization should have lamps to indicate the current state of the operation. It should also have indicators for the filler, conveyor, proximity switch, injector and container level;
5. A table of the input devices giving the address (variable name), acronym and description of each input device;

6. A table of the output devices giving the address (variable name), acronym and description of each output device;

7. Description of the operation of each state (Stop, Execute, etc) with respect to PackML standard and how it operates in the context of the filling of the containers;

8. The weekly group diary of the meetings indicating who attended, meeting date, start time and finish time, a discussion of the quality of the contribution of the tasks allocated at the last meeting, tasks allocated to each member to be completed by the next meeting. It is recommended that a different person complete the diary each week. The completed diary over all the weeks will be used to resolve any conflicts that arise and consequently every member of the group must receive a copy of the current week's diary entry and this should be ratified at the next meeting;

9. The completion of each of the Feedback Fruit surveys by the specified dates. These surveys will be used to determine individual's contribution to the assignment and will be used to calculate the individual's mark for the assignment once the group's assignment mark has been determined;

10. The assignment, including an image of the CODESYS program must be uploaded to the LMS;

11. A USB drive (which can be collected after the assignment has been marked) that contains the program in a form where it can be used to run the program using CODESYS.

Notes:

1. Installation

   - You may use as few or as many instruments as you like, as well as others if the types of instruments listed are not adequate for your chosen control philosophy;
   - For any instrumentation that requires specific dimensional information (i.e. pipe diameter for a flow meter etc) state your chosen dimensions, using information where available or make an assumption and describe why you chose that value;
   - Assume mounting points are available for all instruments at the points you want to put them in the process;
   - Assume 240VAC and 24VDC power supplies are available wherever needed;
   - It is not necessary to establish a communications protocol or use pack.tags

2. Groups

   - You have been allocated to a group to work on this assignment. You cannot change the membership of the group.
   - Attendance at group meetings is compulsory. A penalty of 10% will apply to any individual who misses more than one meeting without sufficient cause verified by the School Student Office in the form of Special Consideration
   - The purpose of the group is to assist the individual in finding relevant, quality sources of information and through discussion among the group come to an understanding and agreement on how to complete the assignment.

- The group is responsible for including all members of the group in the meetings. This means that all members must be contacted and a communication method established. A time and place must be determined so that all members can attend meetings.
- **Warning:** If a member allocated to a group is not included in communications, then unless there is a satisfactory reason (this needs to be confirmed by the unit-coordinator), all members of the group will be penalised 10%.
- You are expected to meet weekly and keep a group diary from the time that the assignment has been allocated.
- The expected average weekly effort on this assignment is 5 hours each person. This includes the time for meetings and discussion, literature search and information assimilation.
- You can meet using any form of communication you prefer: F2F, Wechat, Line, Teams, Zoom etc.

3. References

   - All sources must be declared using the IEEE format for referencing and must be in the english language. Referring to information from the internet requires the url and access date. Referring to the lecture notes is not appropriate for this assessment.

4. The report

   - The quality of the writing is expected to be at a high standard, showing both a command of english (spelling, punctuation and vocabulary) and an understanding of the technical language appropriate to the field.
   - All material appearing in the report that is not the group's original work must be appropriately referenced and a consistent citation style must be used throughout the document.
   - The report will be judged on the following criteria:
     - Clarity & conciseness of the explanations and diagrams;
     - Use & relevance of theories & frameworks;
     - Insight & synthesis of the information collected;
     - Use of sound logical argument
   - The program will be judged on the following items
     - Ability to easily understand the function of the program
     - The ease of changing the program in the event of customer changes
     - Readability of the program
     - The overall operation of the program

5. Feedback-fruits The performance feedback software Feedback-fruits will be used to monitor issues within a group and to make adjustments of the Group Marks to the Individual Marks. There will be 3 surveys through out the semester. The last survey due at the time of submission of the assignment is the main tool used to make adjustments. However, any adjustments require that there is a consistent message coming from all 3 surveys.